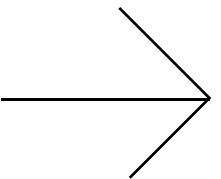




Presented by Samphors Suos

06/19/2026

SpaceX Falcon 9 First Stage Landing Prediction





Executive Summary

Business Problem

SpaceX can significantly reduce launch costs by reusing Falcon 9 first-stage boosters.

Objective

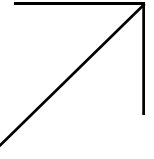
Predict whether the Falcon 9 first stage will successfully land.

Methods

- Data Collection
- Data Wrangling
- Exploratory Data Analysis (EDA)
- Interactive Visual Analytics
- Machine Learning Classification

Key Findings

- Launch site influences landing success.
- Payload mass affects success rates.
- Certain boosters have higher success rates.
- Machine learning models can predict landing outcomes effectively.



Background

SpaceX advertises Falcon 9 launches at approximately \$62 million, much lower than competitors because boosters can be reused.

Problem Statement

Can we predict whether the first stage will successfully land based on launch characteristics?

Project Goal

Build a classification model to predict landing outcomes.

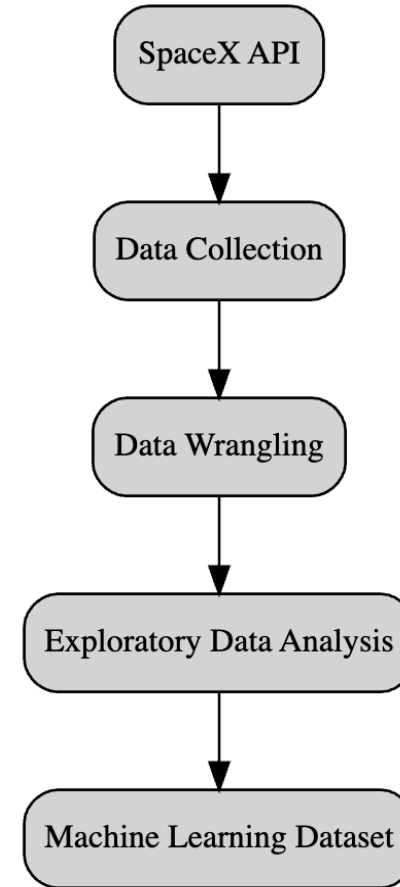
Data Collection Methodology

Sources

- SpaceX REST API
- Web Scraping from Wikipedia

Data Collected

- Flight Number
- Launch Site
- Booster Version
- Payload Mass
- Orbit
- Landing Outcome



Data Wrangling Methodology

Tasks Performed

- Removed missing values
- Standardized landing outcomes
- Created binary target variable
- Converted categorical variables

Tools Used

- Pandas
- NumPy

In [4]:

```
df.isnull().sum()/len(df)*100
```

Out[4]:

FlightNumber	0.000000
Date	0.000000
BoosterVersion	0.000000
PayloadMass	0.000000
Orbit	0.000000
LaunchSite	0.000000
Outcome	0.000000
Flights	0.000000
GridFins	0.000000
Reused	0.000000
Legs	0.000000
LandingPad	28.888889
Block	0.000000
ReusedCount	0.000000
Serial	0.000000
Longitude	0.000000
Latitude	0.000000
dtype:	float64

Identify which columns are numerical and cat

Exploratory Data Analysis Methodology Techniques

SQL Analysis

Scatter Plots

Bar Charts

Correlation Analysis

Questions Investigated

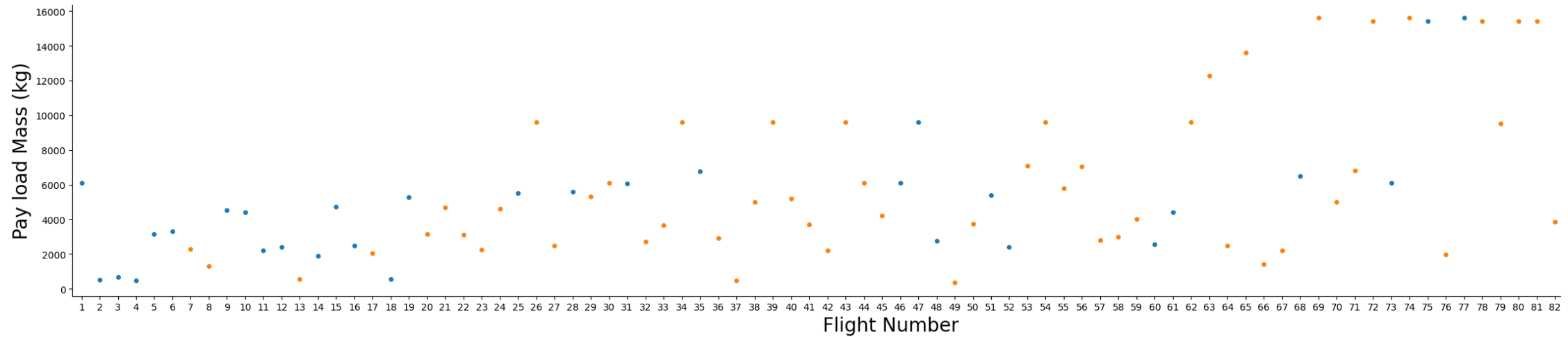
Does payload mass
affect landing success?

Which launch site has
the highest success
rate?

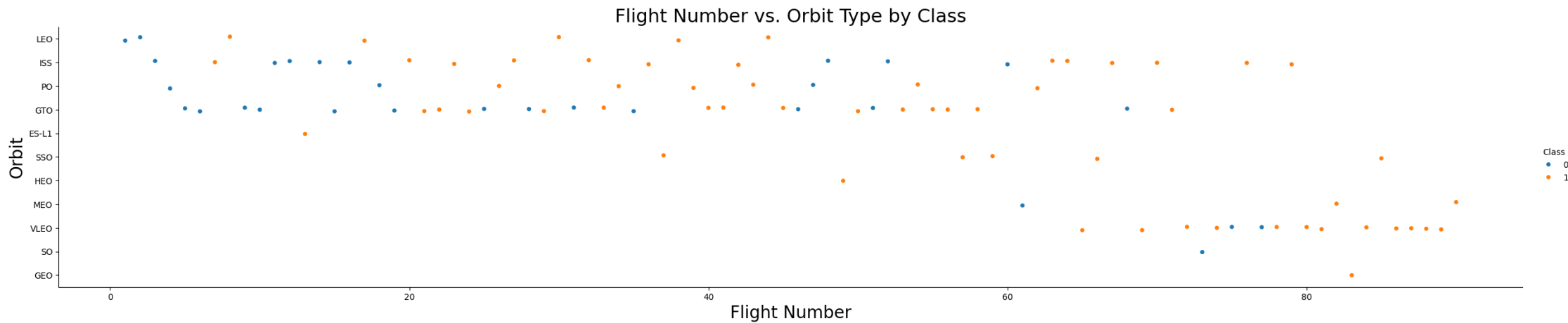
Which orbit performs
best?

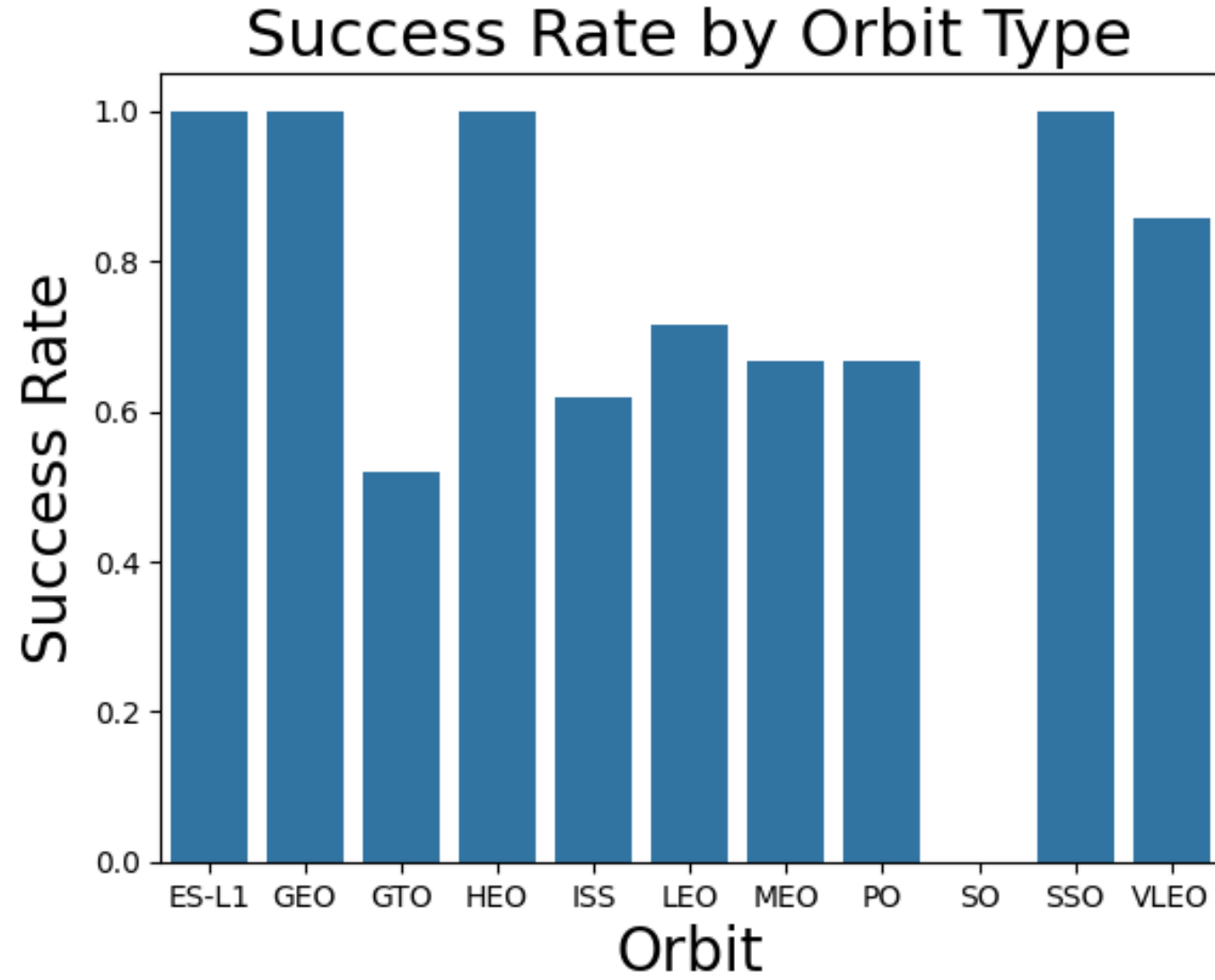


Payload Mass vs. Flight Number by Launch Outcome



Flight Number vs. Orbit Type by Landing Outcome





Success
Rate Across
Different
Orbit Types

EDA with SQL Results

mission_outcomes_summary = %sql SELECT Mission_Outcome, COUNT(Mission_Outcome) AS Total_Count FROM SPACEXTABLE GROUP BY Mission_Outcome; display(mission_outcomes_summary)

...

* sqlite:///my_data1.db

Done.

Mission_Outcome	Total_Count
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

failed_drone_ship_2015 = %sql SELECT SUBSTR(Date, 6, 2) AS Month, Landing_Outcome, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE SUBSTR(Date, 0, 2) = '2015'; display(failed_drone_ship_2015)

* sqlite:///my_data1.db

Done.

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

EDA with SQL Results

```
landing_outcomes_ranked = %sql SELECT Landing_Outcome, COUNT(Landing_Outcome) AS Outcome_Count FROM SPACEXTABLE WHERE Date BETWEEN  
display(landing_outcomes_ranked)
```

```
* sqlite:///my_data1.db
```

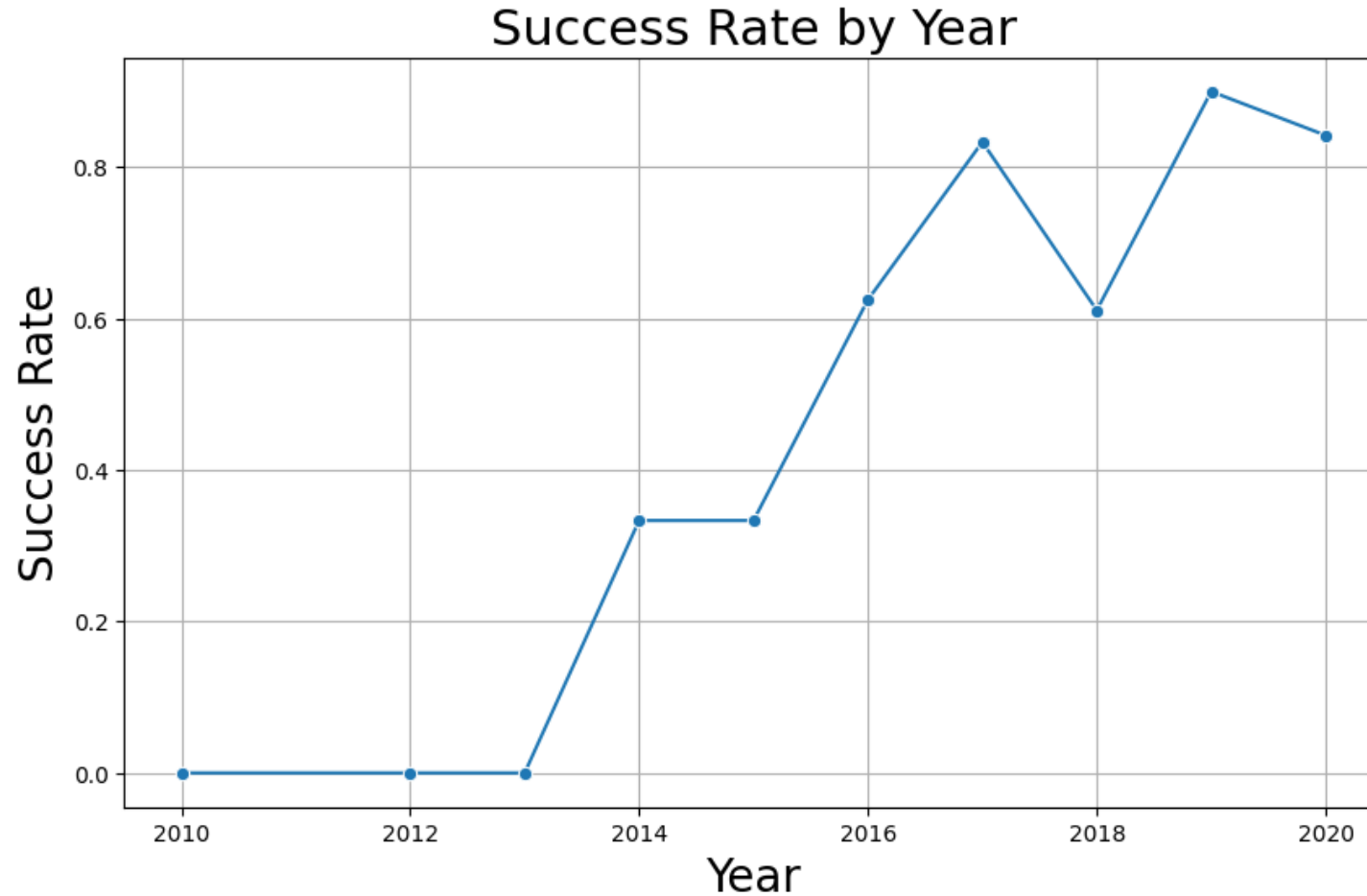
Done.

Landing_Outcome	Outcome_Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Overall SQL Analysis Findings

- SpaceX achieved a very high mission success rate.
- Early missions focused primarily on payload delivery rather than booster recovery.
- Booster landing technology improved substantially over time.
- Successful landings became increasingly common, supporting SpaceX's reusable rocket strategy.

Yearly Launch Success Rate



Predictive Analysis Methodology

Models Evaluated

- Logistic Regression
- SVM
- Decision Tree
- K-Nearest Neighbors

Machine Learning Workflow



Classification Results

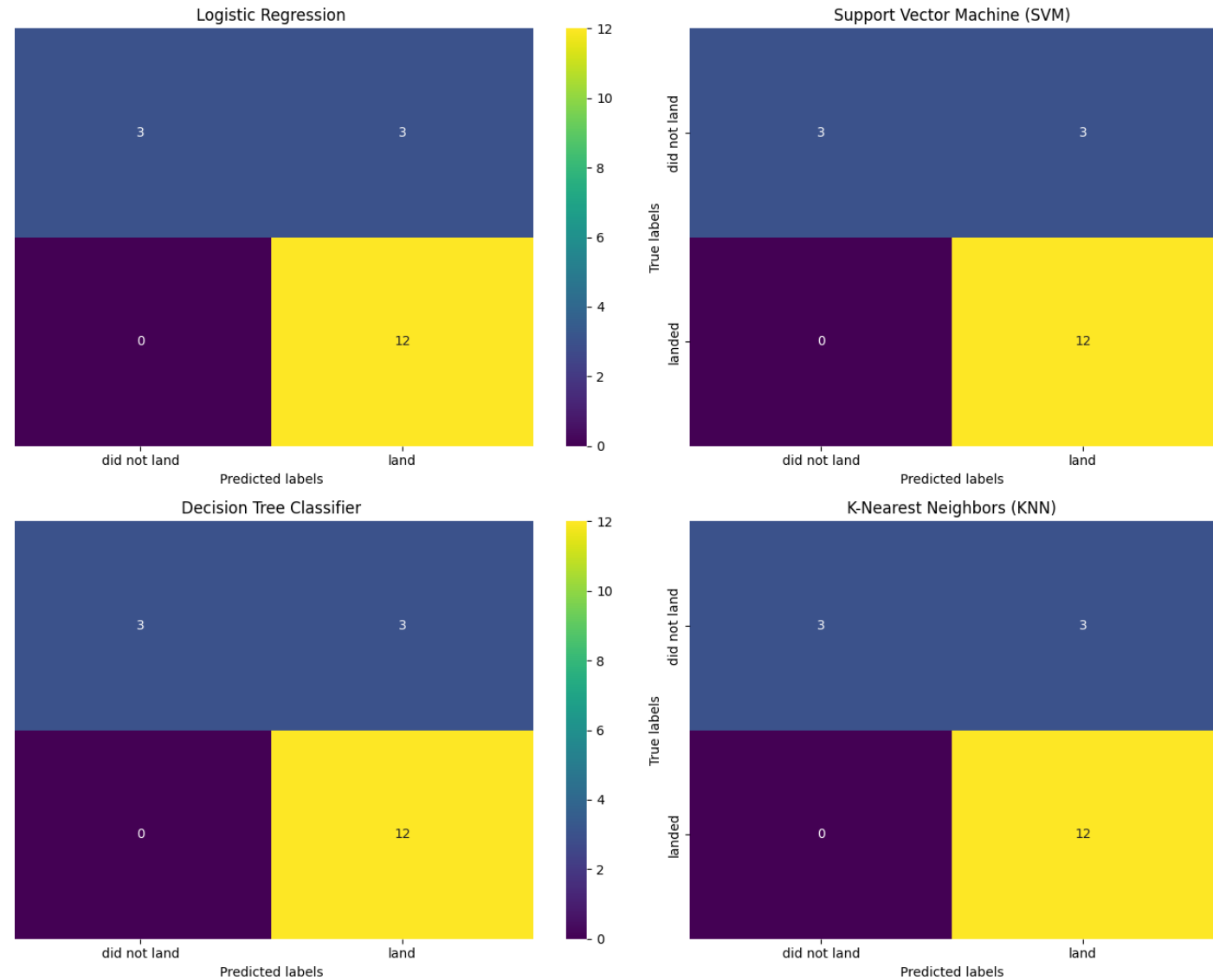
	Model	Accuracy
0	Logistic Regression	83.33%
1	SVM	83.33%
2	Decision Tree	83.33%
3	KNN	83.33%

All four models

Logistic Regression, Support Vector Machine, Decision Tree Classifier, and K-Nearest Neighbors

Performed equally well on the test set with an accuracy of 0.8333

Confusion Matrices for Classification Models

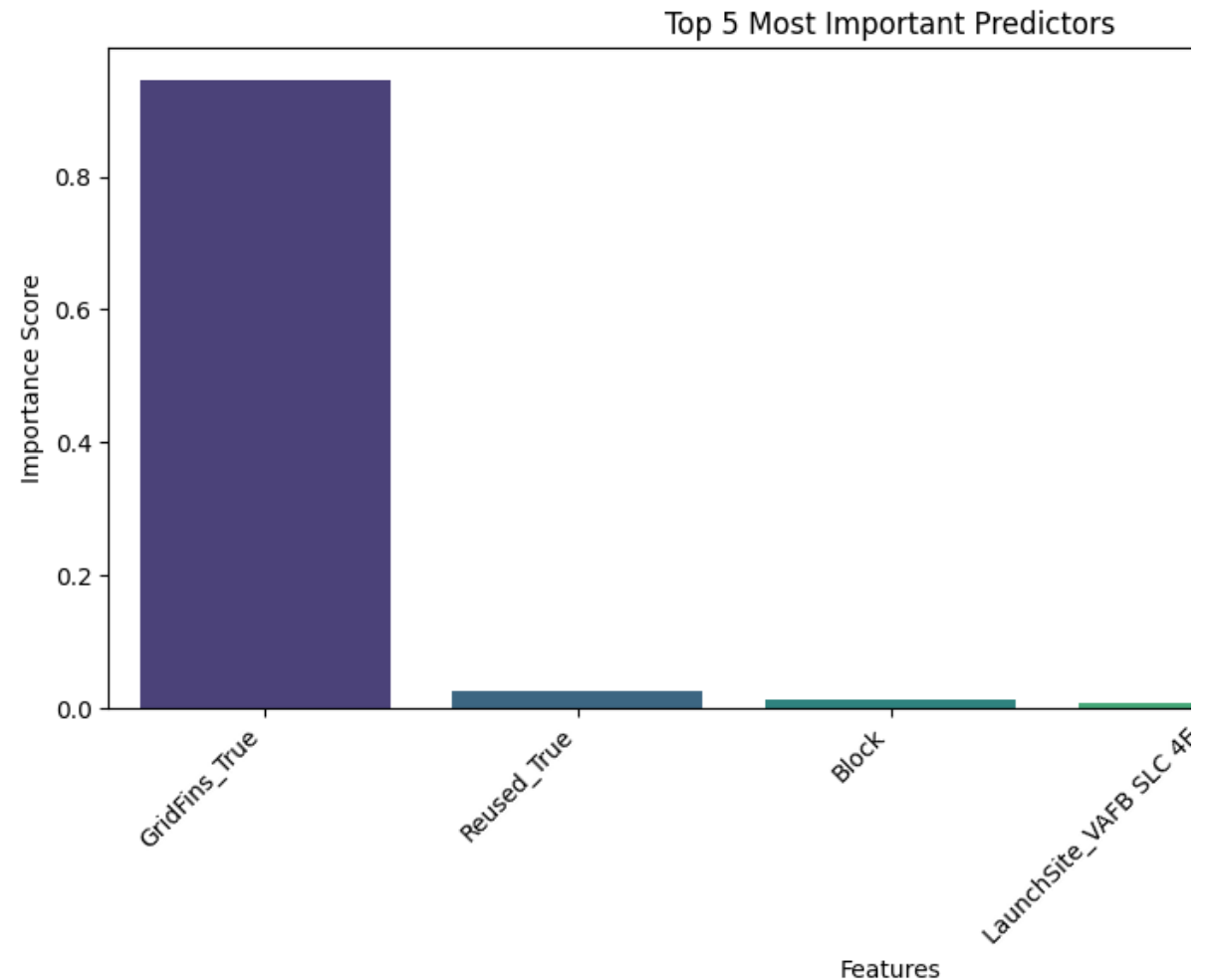


Feature Importance / Insights

Important Predictors:

Payload Mass
Orbit_GTO
Orbit_ISS
Orbit_LEO
Flight Number

Key Insight: Landing success is significantly influenced by the Payload Mass, the intended Orbit type (particularly GTO, ISS, and LEO), and the Flight Number, suggesting that characteristics of the mission and the payload are critical determinants of success.



Conclusion

Objectives Achieved ✓ Collected launch data ✓
Performed data wrangling ✓ Conducted EDA ✓
Built interactive visualizations ✓ Developed
predictive models Main Findings Landing success
improved over time. Payload mass and orbit
influence outcomes. Machine learning can
accurately predict landing success.

Landing success is significantly influenced by
the PayloadMass, the intended Orbit type
(particularly GTO, ISS, and LEO), and
the FlightNumber, suggesting that characteristics
of the mission and the payload are critical
determinants of success.



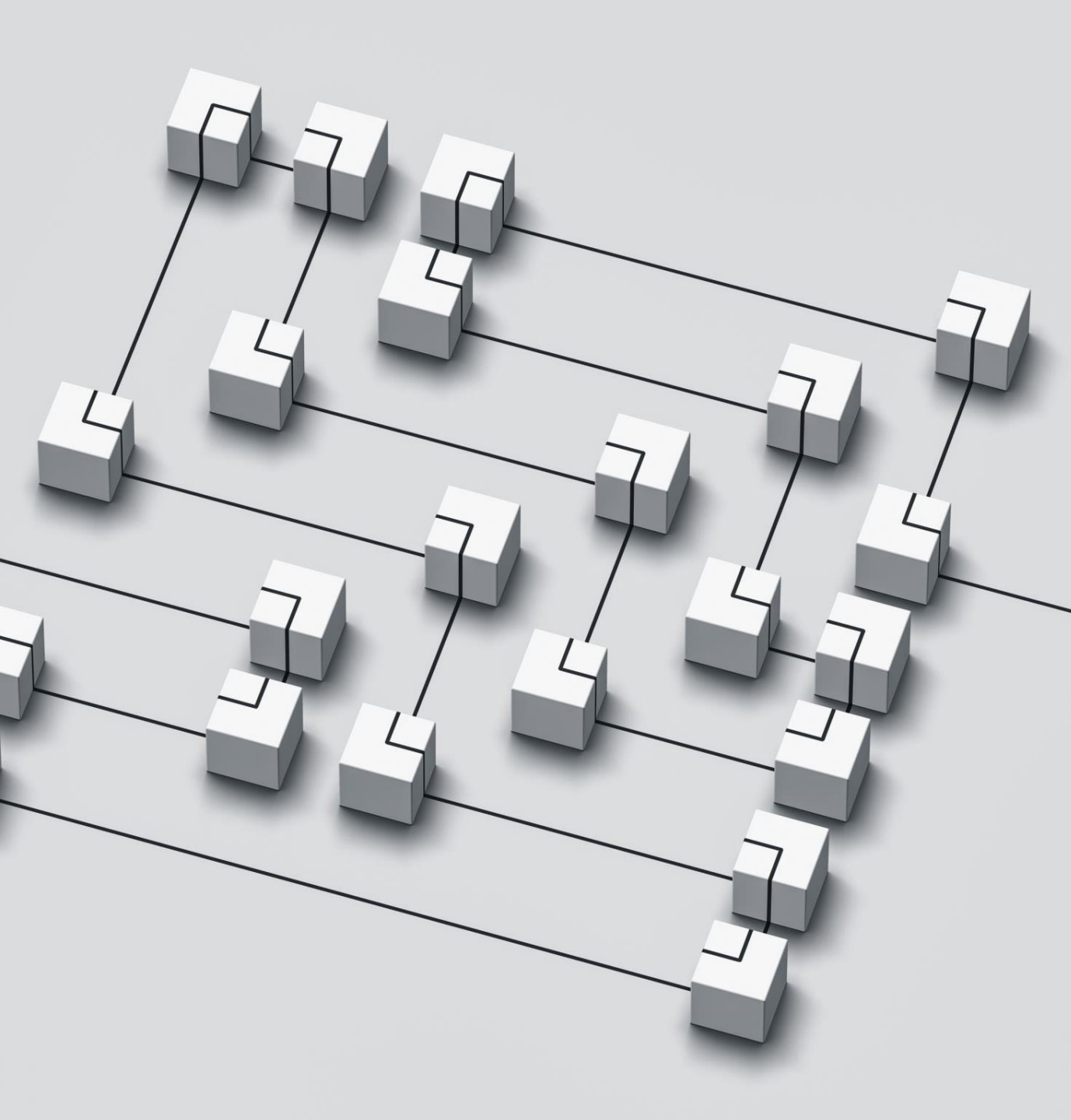
Future Work

Possible Improvements:

- Collect additional launch data.
- Use ensemble methods:
- Random Forest
- XG Boost

Deploy prediction model as a web application.





GitHub Repository & References

GitHub Repository

<https://github.com/phorsya2004-a11y/Final-Project-IBM->

References

- SpaceX API
- Wikipedia Launch Records
- IBM Skills Network Labs



Q&A

Panel discussion session